Proof of $C(n,k) = C(n,n-k)$ by Bijection Principle

- Suppose *S* is a set with n distinct elements.
  Every subset *A* of *S* with $k \leq n$ elements corresponds to the subset *A*'=$S - A$ of *S* with $n-k$ elements.
  Therefore, the function that maps *A* to *A*' is a bijection
  Consequently, the number of subsets of *S* with k elements is the same as the number of subsets of S with $n-k$ elements. The former can be counted by $C(n,k)$ and the latter by $C(n,n-k)$.

Consequently, $C(n,k) = C(n,n-k)$.